

TE Curve Verification Pipeline Familywise ONNX Report - wave4_2_quantile_p10_p50_p90

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report evaluates exported ONNX models from the dataset input-mode retraining program. Each dataset/input-mode section uses dataset-matched held-out test curves and lists the exact model artifacts loaded from `models/`.

Rank-3 temporal ONNX exports are evaluated on the sequence-window test contract stored in each `training_config.snapshot.yaml`, including `sequence_length`, `sequence_stride`, `sequence_target_position`, and `maximum_sequences_per_curve`. The collage pages keep the measured TE trace at the original full-curve resolution; temporal ONNX predictions are overlaid at the evaluated sequence-target angles.

The report is diagnostic and family-specific. It does not replace an official multi-index model-promotion decision.

Output Artifacts

- output directory:
`output/validation_checks/track2_familywise_onnx_report/wave4_2_quantile_p10_p50_p90/2026-07-14-18-14-36__track2_wave4_2_quantile_p10_p50_p90_familywise_onnx_report` ;
- summary YAML:
`output/validation_checks/track2_familywise_onnx_report/wave4_2_quantile_p10_p50_p90/2026-07-14-18-14-36__track2_wave4_2_quantile_p10_p50_p90_familywise_onnx_report/track2_familywise_onnx_report_summary.yaml` ;
- model inventory CSV:
`output/validation_checks/track2_familywise_onnx_report/wave4_2_quantile_p10_p50_p90/2026-07-14-18-14-36__track2_wave4_2_quantile_p10_p50_p90_familywise_onnx_report/model_inventory.csv` ;
- per-curve metrics CSV:
`output/validation_checks/track2_familywise_onnx_report/wave4_2_quantile_p10_p50_p90/2026-07-14-18-14-36__track2_wave4_2_quantile_p10_p50_p90_familywise_onnx_report/per_curve_metrics.csv` .

Simplified Dataset + Setpoints

- dataset: `simplified_dataset` ;
- input mode: `setpoints` ;
- evaluated family: `wave4_2_quantile_p10_p50_p90` ;
- dataset root: `data/simplified_dataset` .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	<code>te_wave4_2_quantile_p10_p50_p90_fw_simplified_setpoints</code>	<code>2026-07-14-11-30-33_te_wave4_2_quantile_p10_p50_p90_fw_simplified_setpoints</code>	<code>simplified_curve_v1</code>
backward	<code>te_wave4_2_quantile_p10_p50_p90_bw_simplified_setpoints</code>	<code>2026-07-14-11-58-31_te_wave4_2_quantile_p10_p50_p90_bw_simplified_setpoints</code>	<code>simplified_curve_v1</code>
global	<code>te_wave4_2_quantile_p10_p50_p90_global_simplified_setpoints</code>	<code>2026-07-14-11-05-29_te_wave4_2_quantile_p10_p50_p90_global_simplified_setpoints</code>	<code>simplified_curve_v1</code>

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	<code>models/simplified_dataset/setpoints/exported/wave4_2_quantile_p10_p50_p90/forward/2026-07-14-11-30-33_te_wave4_2_quantile_p10_p50_p90_fw_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/wave4_2_quantile_p10_p50_p90/forward/2026-07-14-11-30-33_te_wave4_2_quantile_p10_p50_p90_fw_simplified_setpoints/python/curve_aware_harmonic_residual_offset_probe-epoch=098-val_mae=0.00349706.ckpt</code>
backward	<code>models/simplified_dataset/setpoints/exported/wave4_2_quantile_p10_p50_p90/backward/2026-07-14-11-58-31_te_wave4_2_quantile_p10_p50_p90_bw_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/wave4_2_quantile_p10_p50_p90/backward/2026-07-14-11-58-31_te_wave4_2_quantile_p10_p50_p90_bw_simplified_setpoints/python/curve_aware_harmonic_residual_offset_probe-epoch=087-val_mae=0.00355108.ckpt</code>
global	<code>models/simplified_dataset/setpoints/exported/wave4_2_quantile_p10_p50_p90/global/2026-07-14-11-05-29_te_wave4_2_quantile_p10_p50_p90_global_simplified_setpoints/onnx/model.onnx</code>	<code>models/simplified_dataset/setpoints/exported/wave4_2_quantile_p10_p50_p90/global/2026-07-14-11-05-29_te_wave4_2_quantile_p10_p50_p90_global_simplified_setpoints/python/curve_aware_harmonic_residual_offset_probe-epoch=096-val_mae=0.00351589.ckpt</code>

Aggregate Metrics

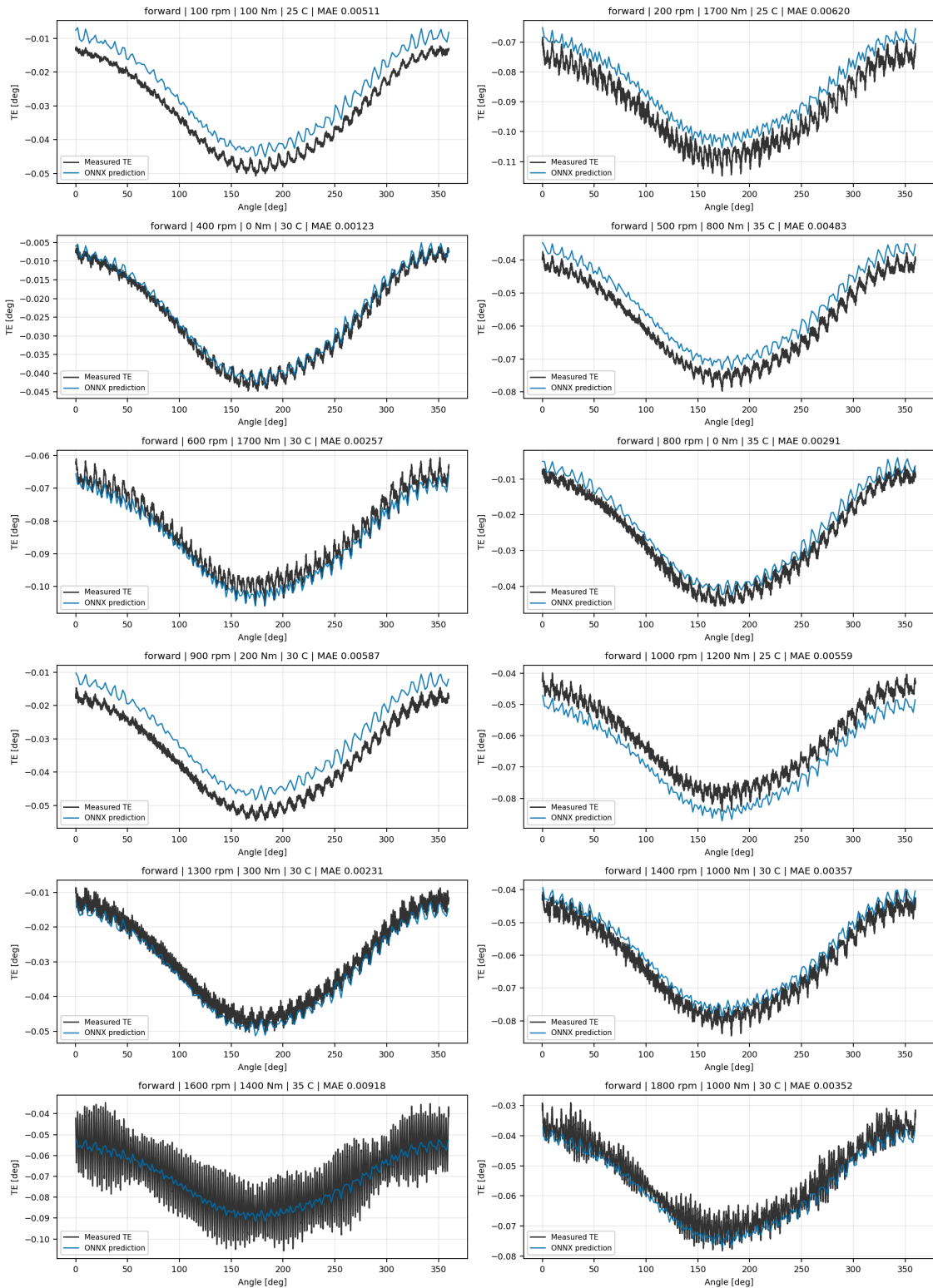
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	97	0.003390	0.003653	7.944	14.140
backward	97	0.003476	0.003786	8.072	15.010
global	194	0.003378	0.003659	7.876	15.980

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	0.000287	0.002984	0.001245	0.003167
backward	0.000280	0.002819	0.001513	0.003509
global	0.000684	0.002850	0.001355	0.003481

Forward 12-Curve Page

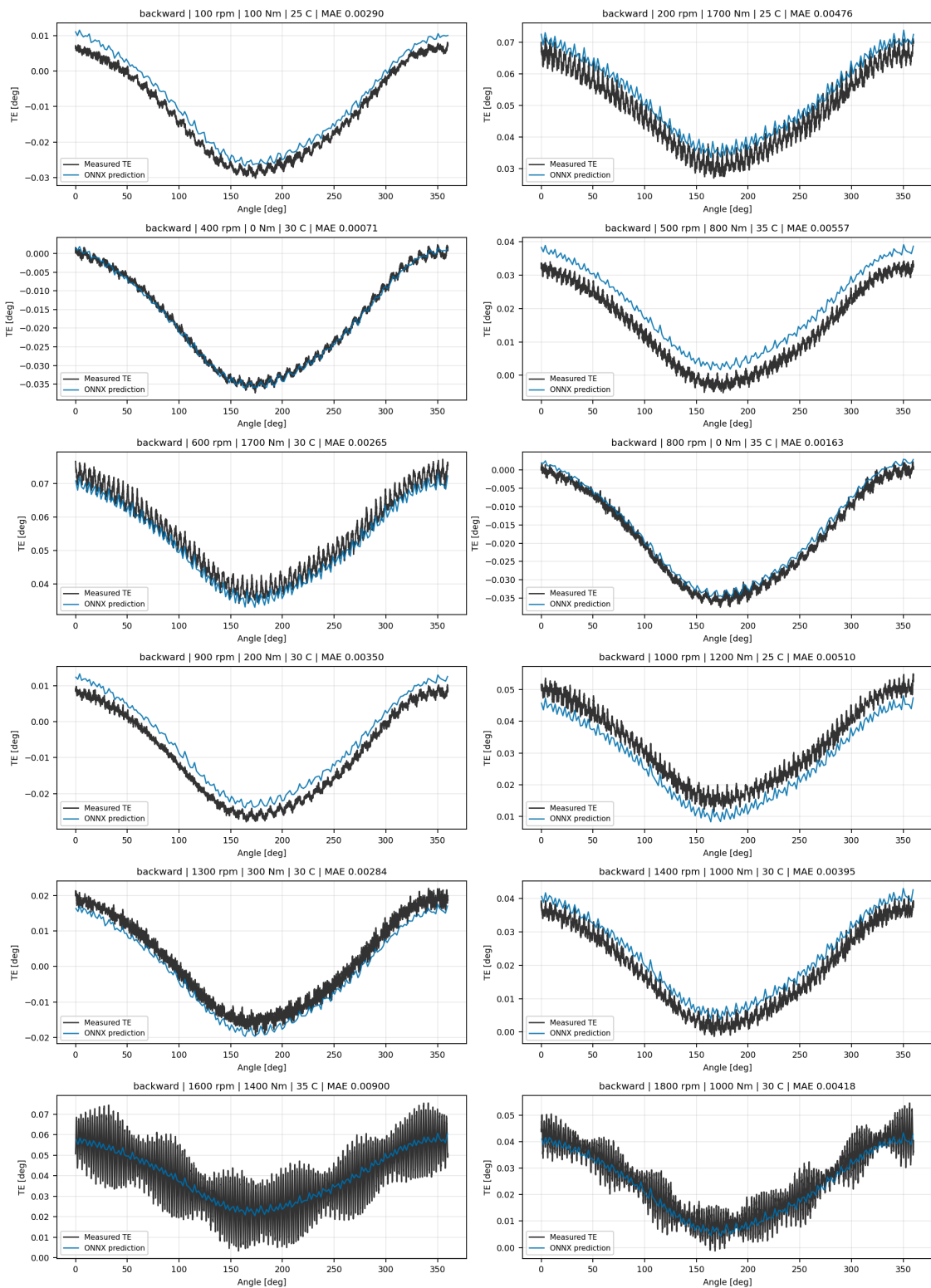
Simplified Dataset + Setpoints | wave4_2_quantile_p10_p50_p90 | forward



simplified_dataset_setpoints forward 12-curve collage

Backward 12-Curve Page

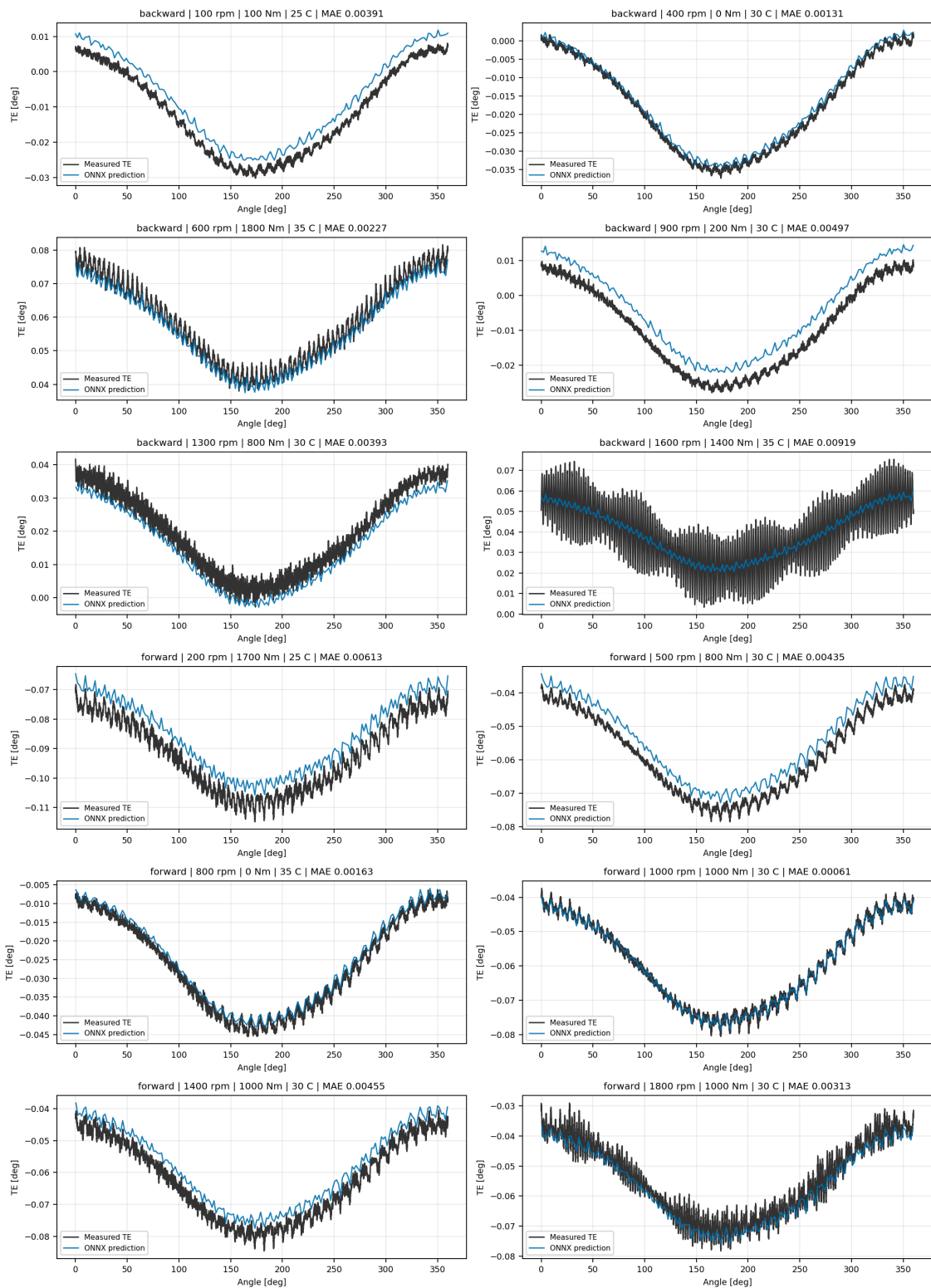
Simplified Dataset + Setpoints | wave4_2_quantile_p10_p50_p90 | backward



simplified_dataset_setpoints backward 12-curve collage

Global 12-Curve Page

Simplified Dataset + Setpoints | wave4_2_quantile_p10_p50_p90 | global



simplified_dataset_setpoints global 12-curve collage

Polished Dataset + Setpoints

- dataset: polished_dataset ;
- input mode: setpoints ;
- evaluated family: wave4_2_quantile_p10_p50_p90 ;
- dataset root: data/polished_dataset .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	te_wave4_2_quantile_p10_p50_p90_fw_polished_setpoints	2026-07-14-13-15-27__te_wave4_2_quantile_p10_p50_p90_fw_polished_setpoints	polished_setpoint_curve_v1
backward	te_wave4_2_quantile_p10_p50_p90_bw_polished_setpoints	2026-07-14-13-54-54__te_wave4_2_quantile_p10_p50_p90_bw_polished_setpoints	polished_setpoint_curve_v1
global	te_wave4_2_quantile_p10_p50_p90_global_polished_setpoints	2026-07-14-12-41-50__te_wave4_2_quantile_p10_p50_p90_global_polished_setpoints	polished_setpoint_curve_v1

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	models/polished_dataset/setpoints/exported/wave4_2_quantile_p10_p50_p90/forward/2026-07-14-13-15-27__te_wave4_2_quantile_p10_p50_p90_fw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/wave4_2_quantile_p10_p50_p90/forward/2026-07-14-13-15-27__te_wave4_2_quantile_p10_p50_p90_fw_polished_setpoints/python/curve_aware_harmonic_residual_of_fset_probe-epoch=157-val_mae=0.00180121.ckpt
backward	models/polished_dataset/setpoints/exported/wave4_2_quantile_p10_p50_p90/backward/2026-07-14-13-54-54__te_wave4_2_quantile_p10_p50_p90_bw_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/wave4_2_quantile_p10_p50_p90/backward/2026-07-14-13-54-54__te_wave4_2_quantile_p10_p50_p90_bw_polished_setpoints/python/curve_aware_harmonic_residual_of_fset_probe-epoch=063-val_mae=0.00181729.ckpt
global	models/polished_dataset/setpoints/exported/wave4_2_quantile_p10_p50_p90/global/2026-07-14-12-41-50__te_wave4_2_quantile_p10_p50_p90_global_polished_setpoints/onnx/model.onnx	models/polished_dataset/setpoints/exported/wave4_2_quantile_p10_p50_p90/global/2026-07-14-12-41-50__te_wave4_2_quantile_p10_p50_p90_global_polished_setpoints/python/curve_aware_harmonic_residual_offset_probe-epoch=109-val_mae=0.00179474.ckpt

Aggregate Metrics

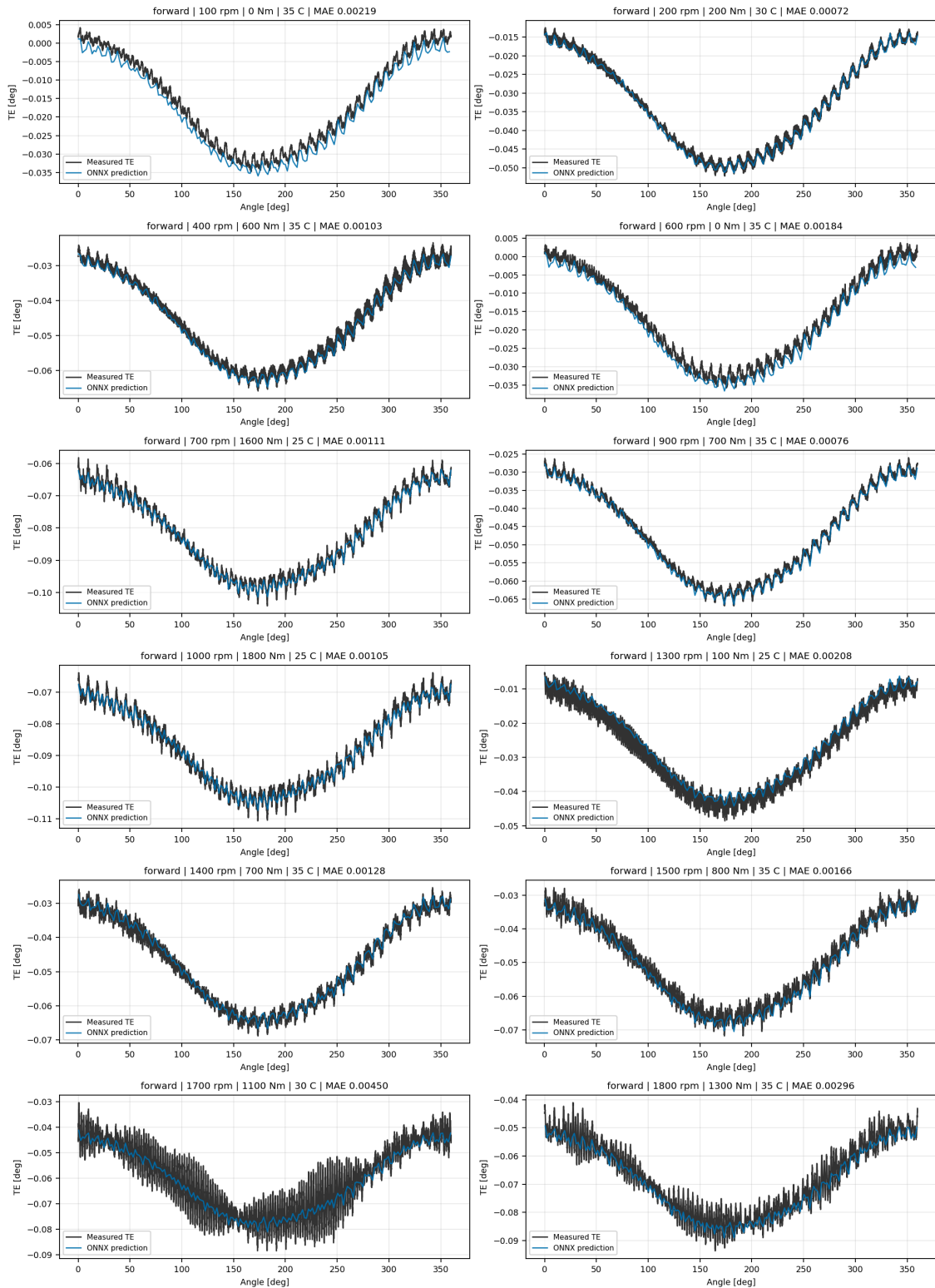
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	100	0.001825	0.002180	3.979	9.470
backward	94	0.002476	0.002904	4.468	10.935
global	194	0.002095	0.002478	4.097	10.805

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	-0.000423	0.000898	0.001506	0.004174
backward	0.000308	0.000936	0.002077	0.005988
global	0.000043	0.000863	0.001766	0.005201

Forward 12-Curve Page

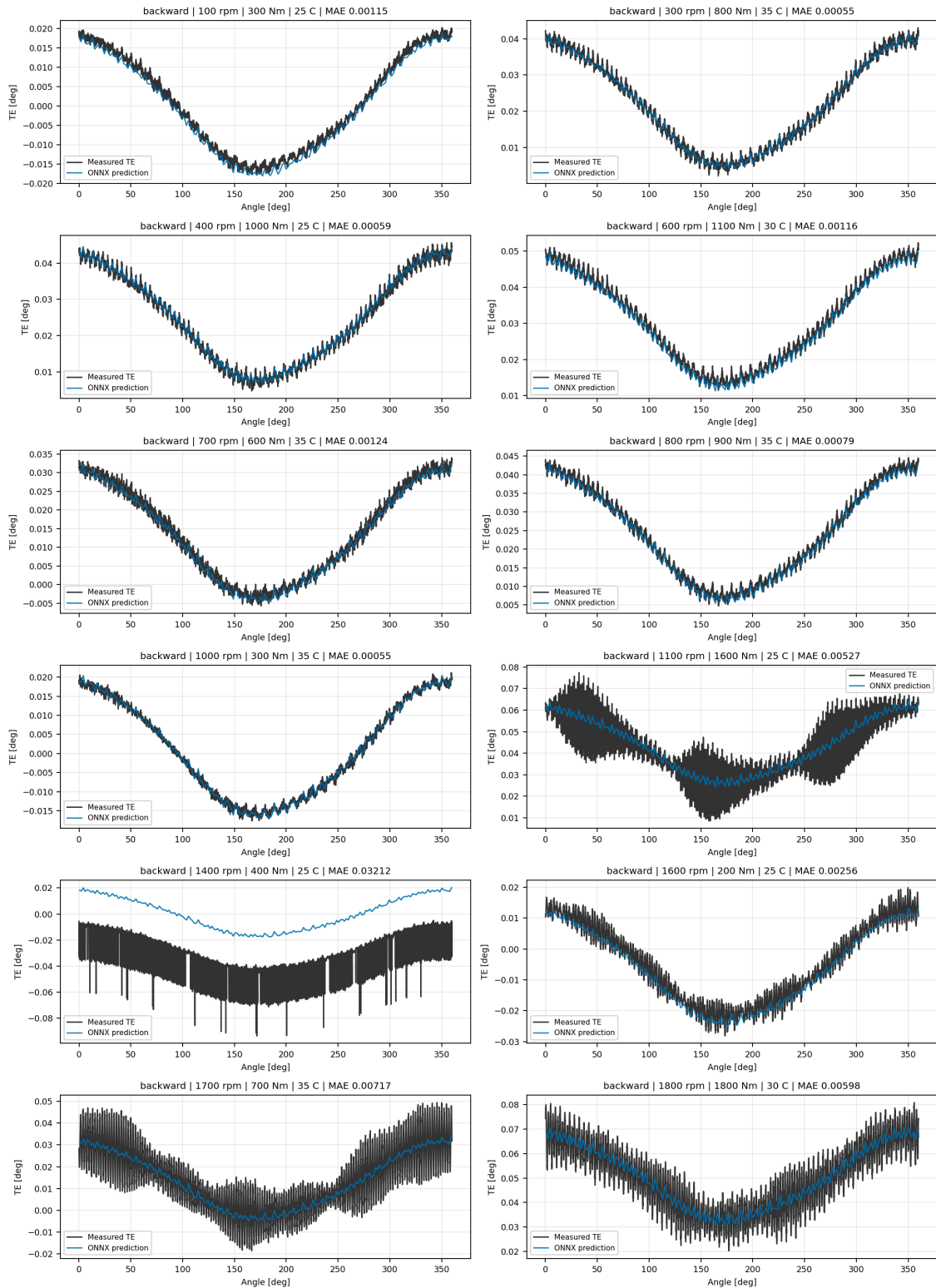
Polished Dataset + Setpoints | wave4_2_quantile_p10_p50_p90 | forward



polished_dataset_setpoints forward 12-curve collage

Backward 12-Curve Page

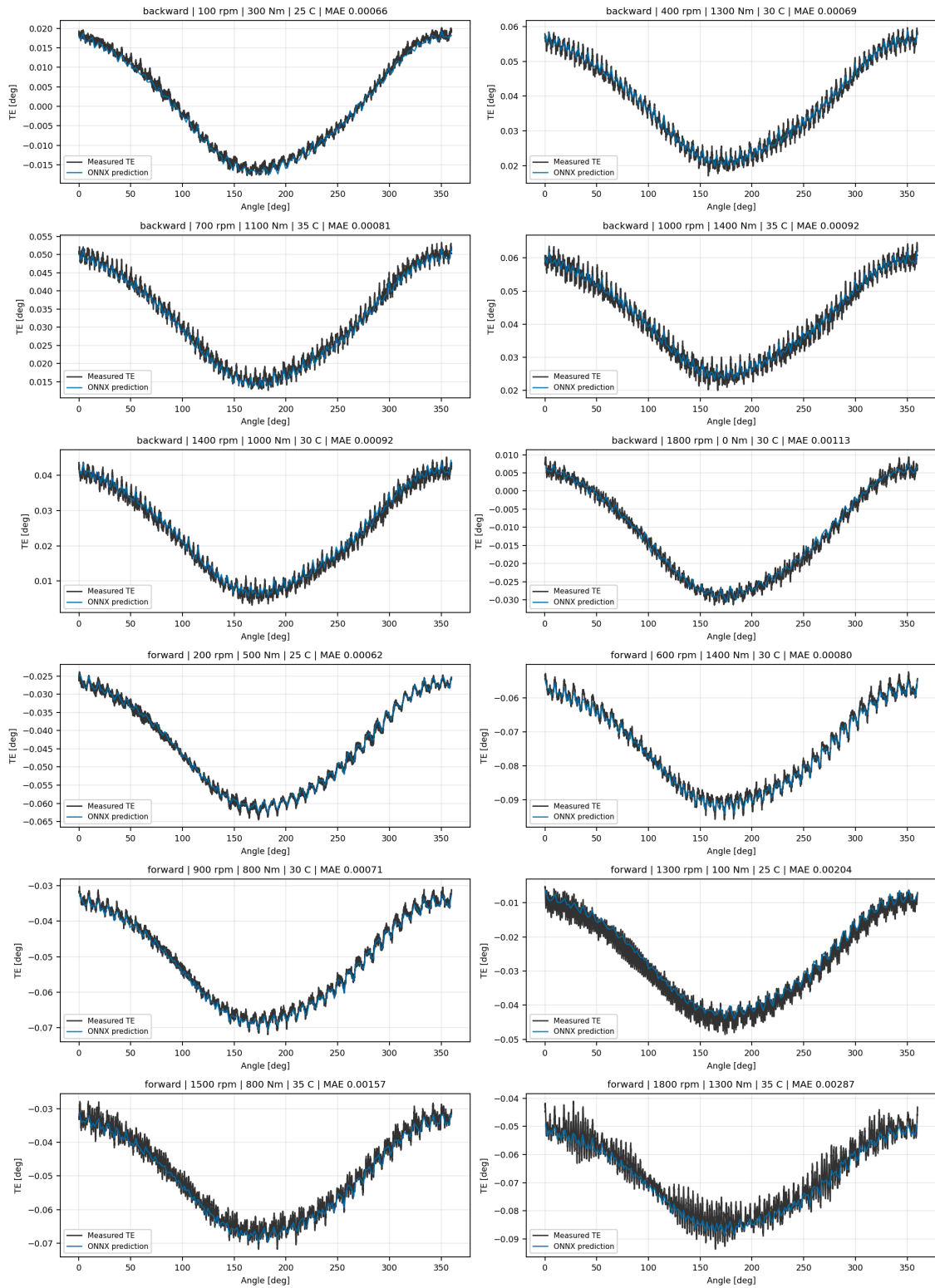
Polished Dataset + Setpoints | wave4_2_quantile_p10_p50_p90 | backward



polished_dataset_setpoints backward 12-curve collage

Global 12-Curve Page

Polished Dataset + Setpoints | wave4_2_quantile_p10_p50_p90 | global



polished_dataset_setpoints global 12-curve collage

Polished Dataset + Actual Values

- dataset: polished_dataset ;
- input mode: actual_values ;
- evaluated family: wave4_2_quantile_p10_p50_p90 ;
- dataset root: data/polished_dataset .

Models Used

Surface	Run Name	Run Instance	Dataset Schema
forward	te_wave4_2_quantile_p10_p50_p90_fw__polished_actual_values	2026-07-14-16-15-03__te_wave4_2_quantile_p10_p50_p90_fw__polished_actual_values	polished_point_v1
backward	te_wave4_2_quantile_p10_p50_p90_bw__polished_actual_values	2026-07-14-17-11-13__te_wave4_2_quantile_p10_p50_p90_bw__polished_actual_values	polished_point_v1
global	te_wave4_2_quantile_p10_p50_p90_global__polished_actual_values	2026-07-14-15-32-15__te_wave4_2_quantile_p10_p50_p90_global__polished_actual_values	polished_point_v1

Exact model paths

Surface	ONNX Model Path	Python Model Path
forward	models/polished_dataset/actual_values/exported/wave4_2_quantile_p10_p50_p90/forward/2026-07-14-16-15-03__te_wave4_2_quantile_p10_p50_p90_fw__polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/wave4_2_quantile_p10_p50_p90/forward/2026-07-14-16-15-03__te_wave4_2_quantile_p10_p50_p90_fw__polished_actual_values/python/curve_aware_harmonic_residual_offset_probe-epoch=250-val_mae=0.00176755.ckpt
backward	models/polished_dataset/actual_values/exported/wave4_2_quantile_p10_p50_p90/backward/2026-07-14-17-11-13__te_wave4_2_quantile_p10_p50_p90_bw__polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/wave4_2_quantile_p10_p50_p90/backward/2026-07-14-17-11-13__te_wave4_2_quantile_p10_p50_p90_bw__polished_actual_values/python/curve_aware_harmonic_residual_offset_probe-epoch=123-val_mae=0.00178818.ckpt
global	models/polished_dataset/actual_values/exported/wave4_2_quantile_p10_p50_p90/global/2026-07-14-15-32-15__te_wave4_2_quantile_p10_p50_p90_global__polished_actual_values/onnx/model.onnx	models/polished_dataset/actual_values/exported/wave4_2_quantile_p10_p50_p90/global/2026-07-14-15-32-15__te_wave4_2_quantile_p10_p50_p90_global__polished_actual_values/python/curve_aware_harmonic_residual_offset_probe-epoch=180-val_mae=0.00177392.ckpt

Aggregate Metrics

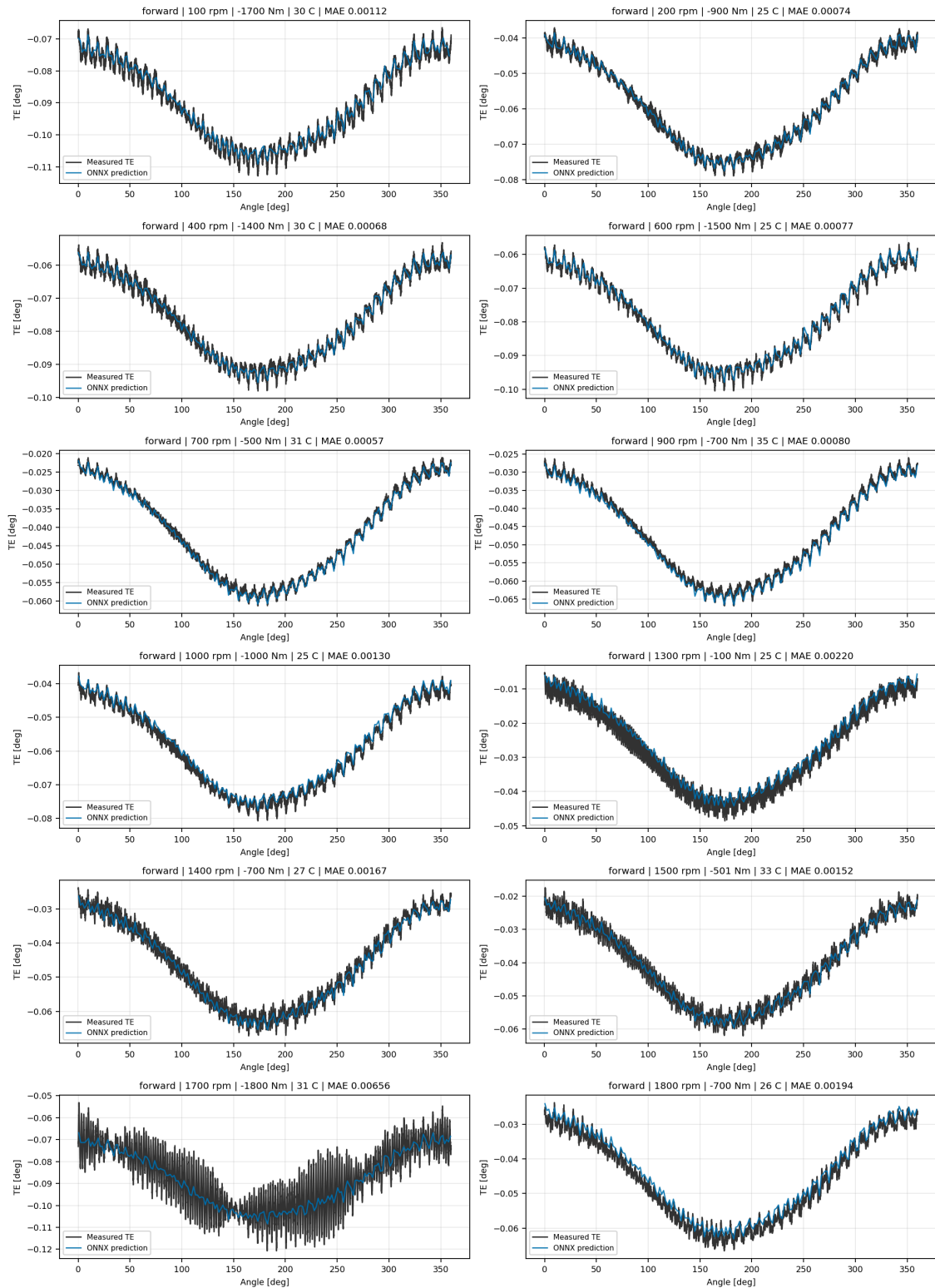
Surface	Curves	MAE[deg]	RMSE[deg]	Mean Error [%]	P95 Error[%]
forward	100	0.001710	0.002057	3.698	9.059
backward	94	0.002179	0.002642	4.064	10.740
global	194	0.001934	0.002326	3.872	10.643

Offset And Shape Metrics

Surface	Signed Offset [deg]	Absolute Offset [deg]	Centered MAE [deg]	P2P Error [deg]
forward	-0.000135	0.000705	0.001500	0.003818
backward	0.000090	0.000590	0.002099	0.006195
global	-0.000035	0.000641	0.001772	0.004877

Forward 12-Curve Page

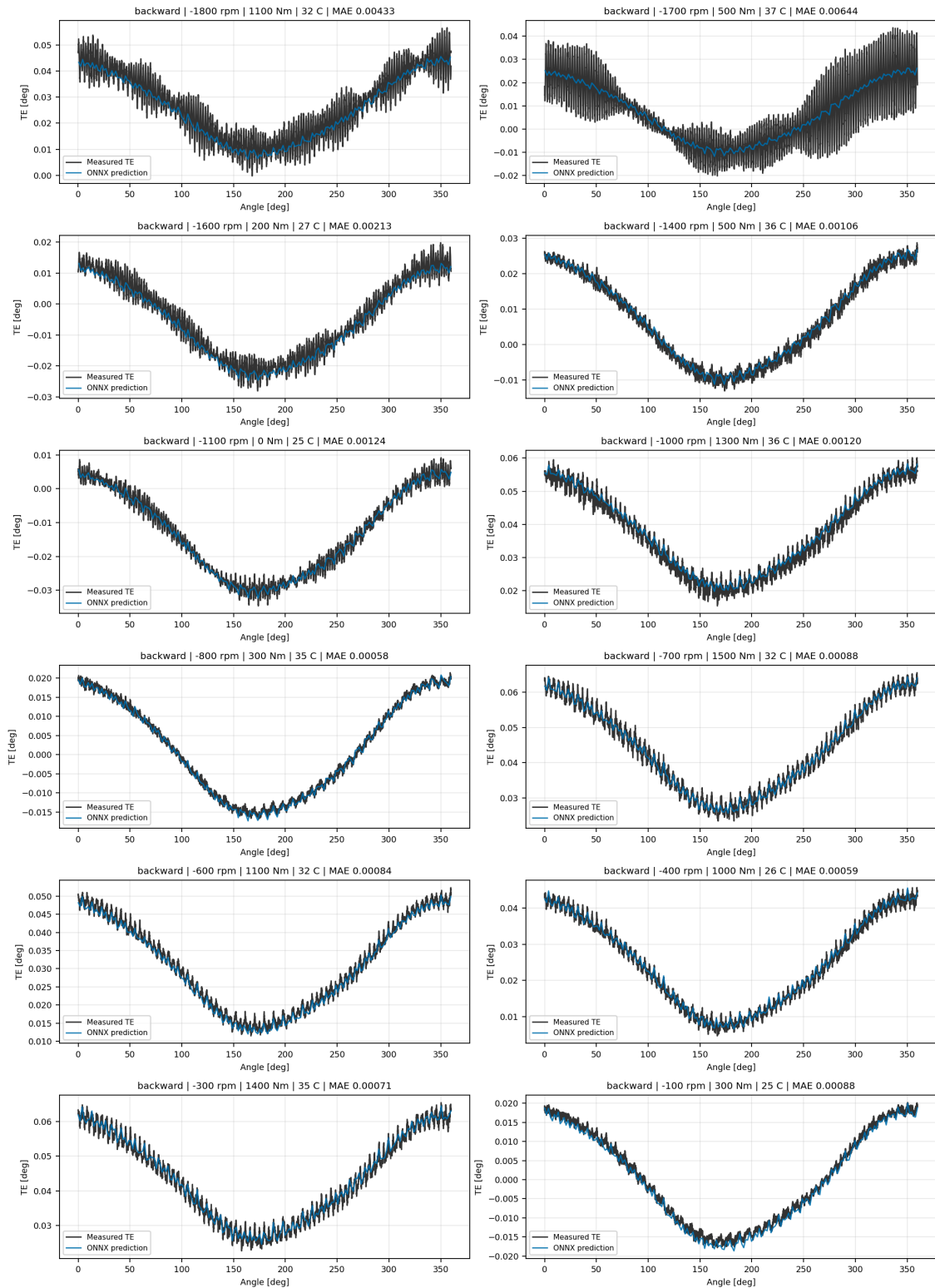
Polished Dataset + Actual Values | wave4_2_quantile_p10_p50_p90 | forward



polished_dataset_actual_values forward 12-curve collage

Backward 12-Curve Page

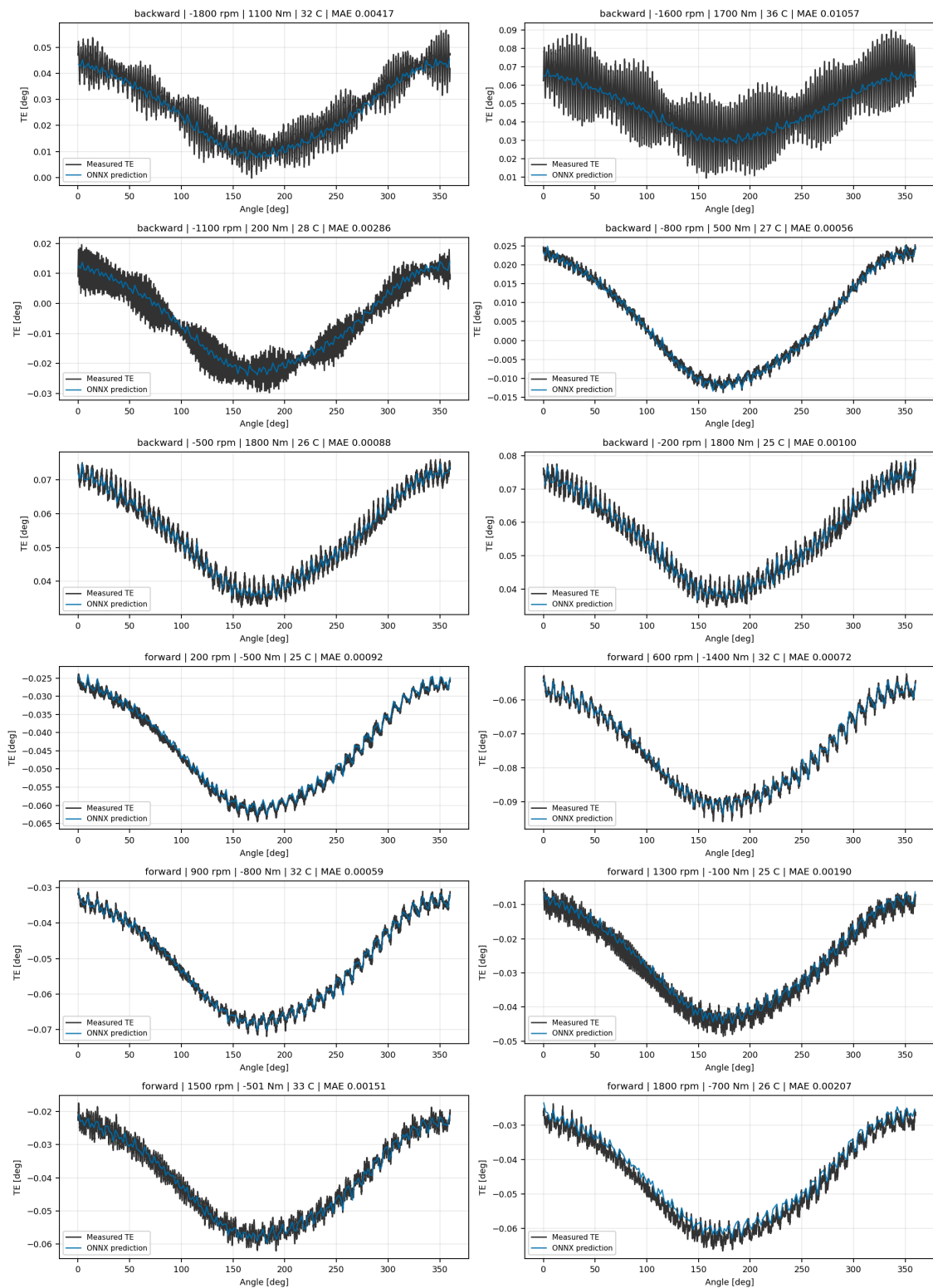
Polished Dataset + Actual Values | wave4_2_quantile_p10_p50_p90 | backward



polished_dataset_actual_values backward 12-curve collage

Global 12-Curve Page

Polished Dataset + Actual Values | wave4_2_quantile_p10_p50_p90 | global



polished_dataset_actual_values global 12-curve collage